

NIKUNJ GARG

Java Backend Developer | BCA Student

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SUMMARY

BCA student focused on Java backend development with Spring Boot 3, PostgreSQL, and Spring Security. Strong foundation in RESTful API design, database transactions, concurrency control, and real-time WebSockets, demonstrated through hackathons, open-source projects, and automated test-driven backends.

TECHNICAL SKILLS

- **Languages & Database:** Java 21, SQL, PostgreSQL
- **Backend Frameworks:** Spring Boot 3, Spring Security, Spring Data JPA, Hibernate, RESTful APIs, WebSockets (STOMP)
- **Security & API Standards:** JWT (Access & Refresh Token Rotation), BCrypt, Bean Validation, OpenAPI/Swagger 3.0
- **DevOps & Tools:** Docker, Docker Compose, Git, GitHub, Maven, Postman, Thunder Client, Cloudinary, Railway
- **Core Engineering:** OOP, Data Structures & Algorithms, Relational Schema Design, Clean Architecture, Unit Testing (JUnit 5)

PROJECTS

AgriSathi AI | Java 21, Spring Boot 3, PostgreSQL, Cloudinary, Swagger [\[GitHub Repo\]](#) • [\[Live Demo\]](#)

- Built the Spring Boot backend for an agricultural advisory platform connecting farmers with buyers during a 30-day hackathon.
- Implemented secure **JWT authentication** with **Role-Based Access Control (RBAC)** to isolate FARMER and BUYER operations.
- Created REST APIs for crop lifecycle tracking, farmer profiles, and crop disease diagnosis (`POST /api/v1/disease/scan`) with Cloudinary media storage.
- Integrated 7-day predictive weather forecasting APIs and documented 15+ endpoints using interactive **OpenAPI/Swagger 3.0**.

DevTinder | Java, Spring Boot, PostgreSQL, WebSockets, Railway [\[GitHub Repo\]](#) • [\[Live Demo\]](#)

- Developed and deployed a teammate matching backend in 72 hours during Devlynix Buildathon 2.0.
- Designed PostgreSQL relational schemas and REST APIs for developer profiles, skill tags, and mutual connection requests.
- Built real-time group and direct messaging using **WebSockets (STOMP protocol)** backed by persistent PostgreSQL chat history.
- Configured production backend deployment on **Railway** with containerized environment variables and health check monitors.

LibroSphere | Java 21, Spring Boot 3, PostgreSQL, Spring Security, JUnit 5 [\[GitHub Repo\]](#)

- Prevented double-booking race conditions on single-copy inventory by enforcing database row locks (`SELECT ... FOR UPDATE`) inside atomic transactions.
- Built an automated FIFO waitlist queue that holds returned books for waiting members with a 48-hour pickup window before restoring general stock.
- Implemented a scheduled overdue fine worker (`@Scheduled`) that runs nightly financial reconciliation and calculates tiered daily penalties.
- Secured APIs with JWT and Refresh Token Rotation (`/auth/refresh`), added composite B-Tree indexes for fast queries, and wrote a **64-test automated test suite** with multi-threaded stress tests (`CountDownLatch`).

EDUCATION

Bachelor of Computer Applications (BCA) — Shri Guru Ram Rai University, Dehradun 2025 – 2028

Cumulative CGPA: 7.48 / 10.0 • **Semester 2:** 7.96 / 10.0

ACHIEVEMENTS & LEADERSHIP

- **Secretary — TechMantra Coding Club:** Promoted from Member (10 months) to Secretary; coordinate student workshops and supported internal rounds for Smart India Hackathon (SIH).
- **Startup Innovation Recognition:** Awarded Certificate of Recognition for presenting an innovative startup idea at the Innovation & Incubation Centre (IIC), SGRR University & UCOST.
- **Java Programming Certification:** Completed Java Programming certification through the IIT Bombay Spoken Tutorial Project.